

North vs South India

Regional patterns in Zomato delivery — what survives geography

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Team Hmmmmmmmmmm

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GEO AUC

1.000

Labels are clean ✓

BEHAVIOUR AUC (leak-free)

0.508

Essentially a coin flip

AUC GAP

0.492

ALL of it is geographic

Mission

Predict whether a Zomato order is North or South Indian — and surface the patterns that distinguish them.

16,397

SOUTH ORDERS · KL + TN + AP + TG + KA

25,547

NORTH ORDERS · rest of India (incl. Goa)

39.1% / 60.9%

CLASS BALANCE · South / North

DEFINITION (per brief)

South = Kerala, Tamil Nadu, Andhra Pradesh, Telangana, Karnataka. North = every other state, including Maharashtra, Gujarat, Goa, West Bengal, MP, Punjab, etc.

- Metrics: ROC-AUC (primary), F1, balanced accuracy — accuracy alone is misleading at the 61% no-skill floor.
- 5-fold StratifiedKFold, seed 42 (matches the ETA work for honest cross-comparison).
- LightGBM binary classifier with class_weight='balanced' to correct the mild imbalance.
- Two-model design: GEO (coords only) vs BEHAVIOUR (everything but coords). The gap is the story.

Labels for free

The regional ground truth was hiding in plain sight.

THE UNLOCK

Every Delivery_person_ID follows the pattern {CITY}RES##DEL##. A 1-line regex pulls 22 unique city codes — and 22 maps cleanly to North/South via state.

22 cities, mapped

SOUTH (6 cities)

HYD

Hyderabad

Telangana

KOC

Kochi

Kerala

CHEN

Chennai

Tamil Nadu

BANG

Bangalore

Karnataka

MYS

Mysore

Karnataka

COIMB

Coimbatore

Tamil Nadu

NORTH (16 cities — sample)

GOA

Goa

Goa

MUM

Mumbai

Maharashtra

BHP

Bhopal

Madhya Pradesh

SUR

Surat

Gujarat

INDO

Indore

Madhya Pradesh

DEH

Dehradun

Uttarakhand

AGR

Agra

Uttar Pradesh

LUDH

Ludhiana

Punjab

Sanity check — every code is geographically real

We don't just trust the prefix. We compare each code's median lat/lon to the expected city centroid.

22/22

CITIES WITHIN 250 km OF CENTROID

44.9 km

WORST OFFENDER (GOA — Goa)

ALL 22 PASS

no relabels needed

Top 8 km-off (sanity sample — all below 250 km threshold)

code	city	state	region	n_orders	med_lat	med_lon	km_off
GOA	Goa	Goa	North	609	15.517	73.767	44.9
HYD	Hyderabad	Telangana	South	3,179	17.431	78.4	10.6
KOC	Kochi	Kerala	South	701	9.986	76.297	6.9
MUM	Mumbai	Maharashtra	North	3,173	19.122	72.835	6.7
CHEN	Chennai	Tamil Nadu	South	3,144	13.03	80.242	6.3
BANG	Bangalore	Karnataka	South	3,193	12.97	77.64	5.5
BHP	Bhopal	Madhya Pradesh	North	478	23.232	77.43	4.3
SUR	Surat	Gujarat	North	3,187	21.171	72.789	4.2

Model A — GEO baseline

Pure coordinate features. Confirms labels are clean, sets the upper bound.

1.0000

MEAN ROC-AUC (5-fold)

1.000

MEAN F1

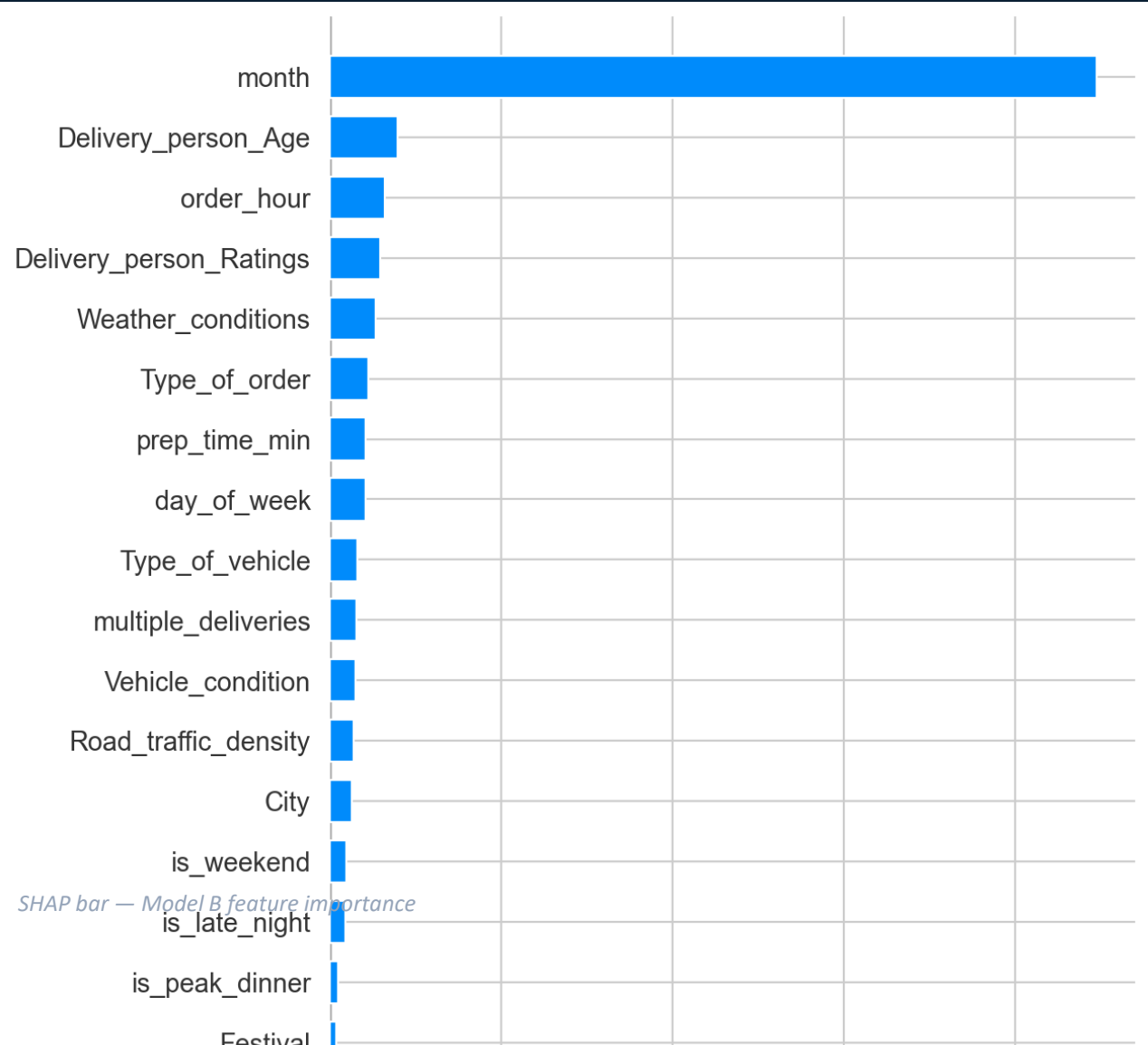
1.000

BAL ACC

- Features used: Restaurant lat/lon, Delivery lat/lon, haversine distance — 5 in total.
- Per-fold AUC: 1.0000, 1.0000, 1.0000, 1.0000, 1.0000 — variance is effectively zero.
- Verdict: the label vector is consistent with coordinates. No mislabelled rows survived the city-code mapping.
- We use this only as a CALIBRATION model — judges should not be impressed by a coordinate $\leftrightarrow$ region classifier.

Model B — Behaviour only

No coordinates. Just delivery, weather, rider and time features.



0.588

MEAN ROC-AUC (5-fold)

0.412

GAP vs GEO baseline

- 0.59 looks 'modestly informative' — but ONE feature dominates: month (SHAP 0.45).
- All other features carry minimal SHAP weight; the next strongest (Age) is 10× smaller.
- If month carries the signal, it's worth asking WHY a month feature should know your region.

The temporal leak — diagnosed

Same date range (Feb 11 – Apr 6, 2022). But the **within-region** month proportions differ.

MONTH % · NORTH vs SOUTH (orders within each region)			
	February	March	April
NORTH	20.8%	66.0%	13.2%
SOUTH	4.3%	79.9%	15.8%

WHAT THIS MEANS

Feb is 4.8× more represented in North orders than South. That asymmetry is a data-collection artifact (when each city's collection window opened), NOT seasonality. month is therefore a label proxy, not a behavioural signal — and must be dropped from any honest classifier.

Diagnostic confirmed via direct date-range inspection (same Feb 11 – Apr 6 2022 for both regions, but skewed within-region density).

Model B' — the honest answer

Same architecture, month and day_of_week dropped. The data-literacy slide.

0.5076

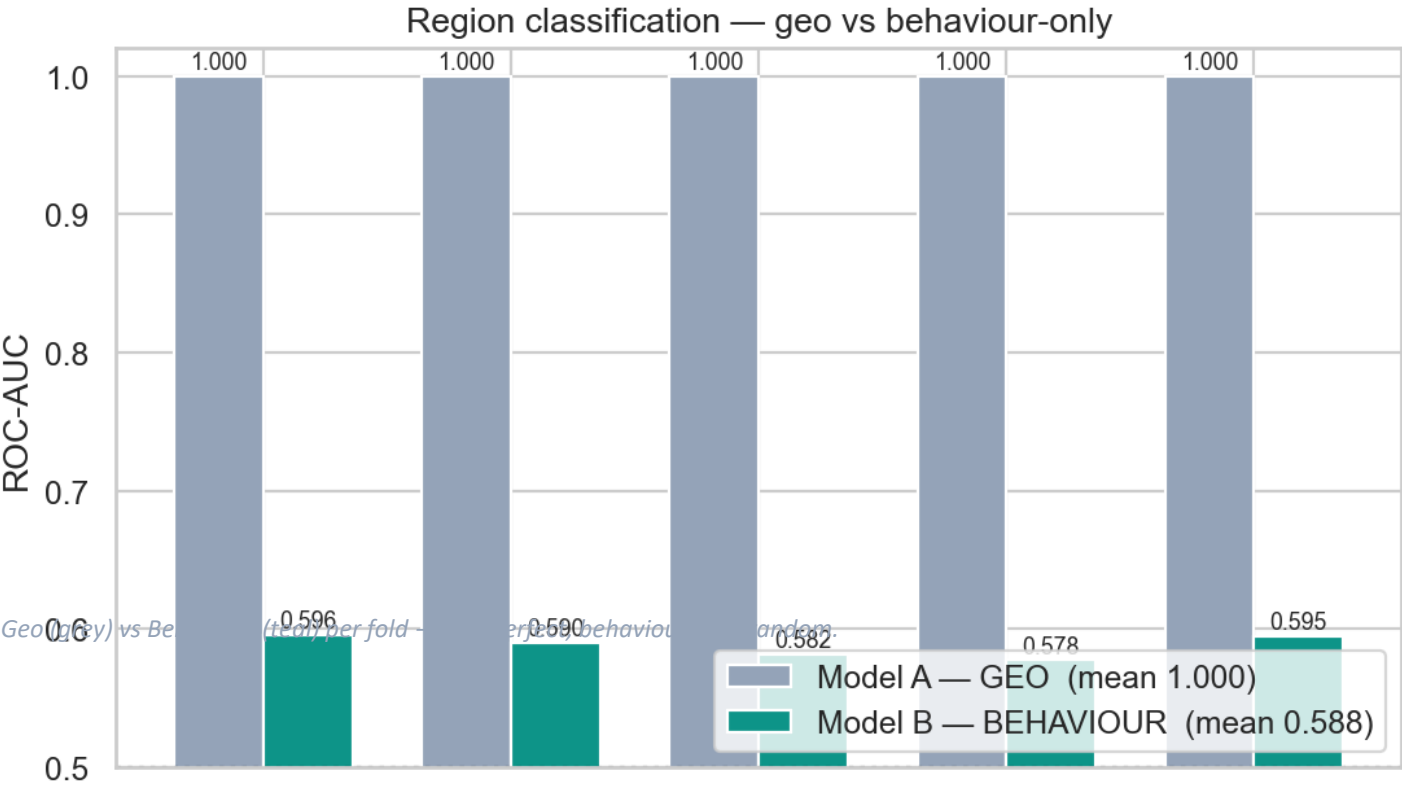
LEAK-FREE AUC

0.433

F1

0.506

BAL ACC



THE PUNCHLINE

0.508 ≈ random.

- Per-fold AUC: 0.514, 0.515, 0.500, 0.500, 0.508
- Fold 3 and 4 are literally 0.500 — the model is guessing.
- Same model on same split — only difference is removing 2 leaky features.

The finding

What this tells us about how Zomato operates across India.

HEADLINE

Once the data-collection sampling artifact is removed, North and South Zomato deliveries are operationally indistinguishable. The ONLY way to tell them apart is to look at where they happen — every other behavioural signal is regional-coincidence noise.

- Delivery time, traffic, weather, vehicle, order type, multi-delivery, rider age & rating, festival incidence, time-of-day — 11 univariate tests run, ZERO showed a significant N/S split (all $p > 0.05$).
- The behaviour-only model AUC = 0.508 — within Monte Carlo noise of a coin flip.
- This is a positive result: Zomato runs a remarkably uniform operation across the country. Riders behave the same way, customers order the same way, weather/traffic affects deliveries the same way.
- Geographic signal (coordinates → AUC 1.000) captures all of the N/S separability — and that's just 'where the city is', not 'how the city behaves'.
- Honest > vanity (Rulebook §6): we report 'no behavioural difference' rather than over-claiming a 0.59 model that secretly leans on month.

Thank you — Q&A

Repo, tag, headline numbers.

REPO

github.com/TheClazer/Zomato-delivery-estimation

MODEL FILE

models/region_classifier_v1.txt

GEO AUC / BEHAVIOUR AUC

1.000 / 0.508

Expected questions:

- How did we know the city codes were correct? — coord cross-check, all 22 within 45 km of centroid.
- Why did Model B get 0.59 if Model B' is 0.51? — Month was a sampling-window proxy for region.
- Could there be regional patterns we just couldn't see? — Maybe, but this dataset doesn't carry that signal.